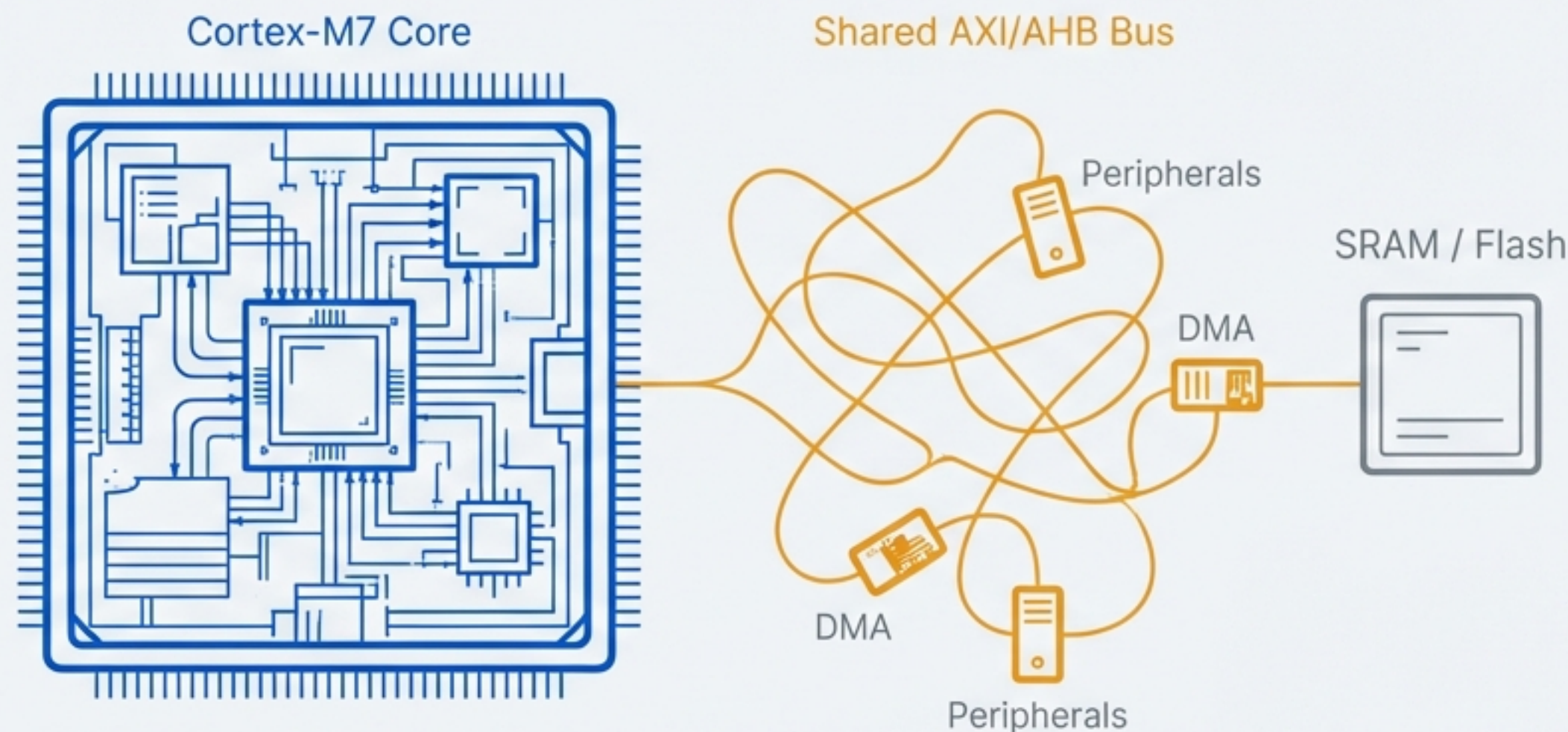


The Performance Bottleneck in High-Speed MCUs

Your Cortex-M7 is Fast. Is Your Memory Keeping Up?



The STM32H7's Cortex-M7 is a high-performance, superscalar processor designed for demanding tasks.

Its biggest performance limiter is often not the core speed, but the time spent waiting for instructions and data.

Standard memory (SRAM, Flash) is accessed through a shared AXI/AHB bus matrix, creating contention and latency.

- CPU, DMA, and other peripherals compete for access.
- This leads to unpredictable delays and wait states, which is unacceptable for real-time applications.

The core question: How do we give the CPU a private, high-speed path to its most critical resources?

The Solution: Tightly-Coupled Memory (TCM)

A Private, Zero-Wait-State Workspace for the CPU



TCM is high-speed SRAM with a dedicated, private connection to the Cortex-M7 core.

This enables memory access at full CPU clock speed, with **zero wait states**. Source: STM32Ref.pdf, Sec 2.1.2

The memory system is split into two specialized regions:

- **ITCM (Instruction TCM):** A 64-bit interface for fetching and executing performance-critical code.
 - Ideal for interrupt service routines and core algorithms. Source: DDI0489B_cortex_m7_trm.pdf, Sec 1.3.6
- **DTCM (Data TCM):** Two 32-bit interfaces (D0TCM & D1TCM) for accessing critical data.
 - Designed for stacks, heaps, and frequently accessed variables. Source: DDI0489B_cortex_m7_trm.pdf, Sec 1.3.6

TCM is enabled via the `CM7\_ITCMCR` and `CM7\_DTCMCR` control registers. Source: DDI0489B_cortex_m7_trm.pdf, Sec 3.3.6

The Architectural Advantage: Access Path is Everything



Tightly-Coupled Memory (TCM)

The Analogy: **A CPU's Private Workbench.**

Access Path: Direct, private bus to the CPU core.
(Source: STM32Ref.pdf, Sec 2.1.2)

Key Characteristics:

- **No bus arbitration** or contention from other masters (like DMA).
- Guaranteed **zero-wait-state** access at CPU clock speed.
- Always treated as **Non-cacheable, Non-shared Normal** memory. (Source: DDI0489B_cortex_m7_trm.pdf, Sec 5.7.1)



Standard SRAM

The Analogy: **A Shared Workshop Floor.**

Access Path: Accessed via the main AXI/AHB bus matrix.
(Source: DDI0489B_cortex_m7_trm.pdf, Sec 1.1)

Key Characteristics:

- Access is **shared** and **arbitrated** between the CPU, DMA, and other peripherals.
- Potential for significant **latency** and unpredictable wait states.

Core Advantages



Extreme Performance: The CPU never stalls waiting for code or data in TCM.



Ultra-Low Latency: Bypassing the system bus matrix drastically reduces access time.



Deterministic Timing: Crucial for hard real-time tasks where predictable execution is non-negotiable.

Practical Application: Where to Use TCM for Maximum Impact

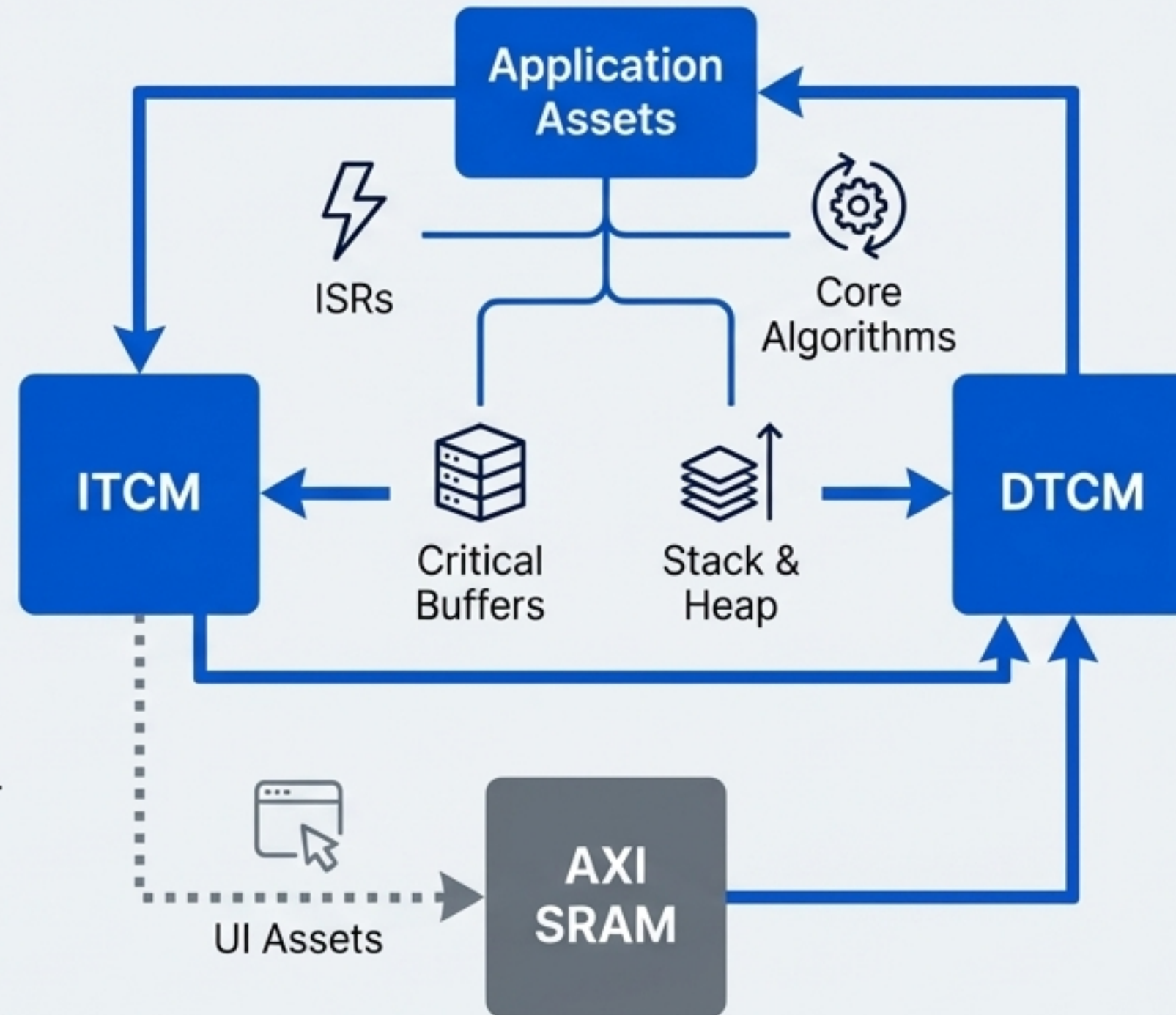
Strategically Place Your Most Time-Sensitive Assets

Recommended Placement

Interrupt Service Routines (ISRs):

Place ISR code in ITCM and associated data/flags in DTCM for the fastest possible interrupt response, leveraging the core's low-latency interrupt handling. (Source: DDI0489B_cortex_m7_trm.pdf, Sec 1.2.5)

Critical Data Buffers: Locate DMA buffers, communication FIFOs, or real-time sensor data in DTCM. The AHBS interface provides high-bandwidth DMA access to TCM. (Source: DDI0489B_cortex_m7_trm.pdf, Sec 5.6)

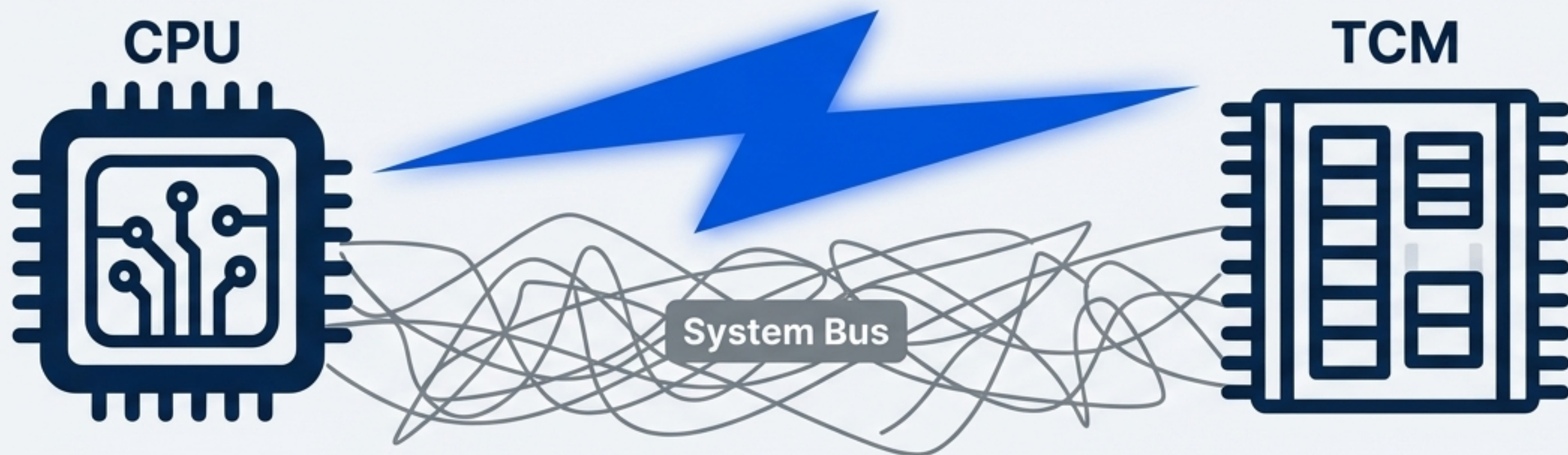


Stack & Heap: Placing the stack and/or heap in DTCM dramatically accelerates function calls, context switching, and dynamic memory allocation. (Source: STM32Ref.pdf, Sec 2.4.1)

Core Algorithms: Move frequently executed, time-critical routines like DSP functions, motor control loops, or protocol stacks into ITCM to ensure they run without memory-induced stalls. (Source: STM32Ref.pdf, Sec 2.4.1)

Unlocking Full Potential: The TCM Takeaway

TCM is the CPU's private, zero-wait-state memory.



Eliminate Bottlenecks

Use TCM to bypass the shared system bus for your most critical operations.

Minimize Latency

The direct connection provides the fastest possible access to code and data.

Guarantee Determinism

Achieve predictable, real-time performance by eliminating contention from DMA and other peripherals.

Strategic use of ITCM and DTCM is not an optimization—it is essential for unlocking the true real-time performance of the STM32H7.